

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-7002-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Structural Design -II (Steel)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Use of relevant IS code is permitted.

1. a) Define characteristic load. What is the use of partial safety factors? How they are different from the factor of safety used in the working stress method?
b) Design a welded end plate connection for a ISMB 400 in grade Fe 410 steel to carry a reaction of 120 kN due to factored loads. The connections has to be made to the flanges of ISHB 300.
2. Calculate the compressive resistance of a compound column consisting ISMB500 with one cover plate 350×20 mm on each flange and having a length of 5m. Assume that the bottom of column is fixed and top is rotation fixed, translation free. Take $F_y = 250 \text{ N/mm}^2$.
3. Explain the steps involved in the design of gantry girder.
4. a) What are the steps involved in the design of plate girders?
b) Determine the moments and forces due to the vertical and horizontal loads acting on a simply-supported gantry girder given the following data.

i) Simply supported span	= 6 m
ii) Crane's wheel centres	= 3.6 m
iii) Self-weight of the girder	= 1.6 k N/m
iv) Maximum crane wheel load	= 220 k N
v) Weight of crab/trolley	= 60 k N
vi) Maximum hook load	= 200 k N

CE-7002-CBGS

PTO

[2]

5.
 - a) Calculate the design compressive load for a column made up of ISHB 350 @ 710.2 N/m and 3.5m high. The column is restrained in direction and position at both the ends. Use steel of grade Fe410.
 - b) Design a steel column using two elements formed together for a factored axial load of 1000kN. The length of the column is 8m and it is hinged at top and fixed at bottom. Use ISMB sections.
6. Design a slab base for a column ISHB 300 @ 577 N/m carrying an axial load of 1200 kN. M20 concrete is to be used for the foundation. Provide welded connection between the column and base plate.
7.
 - a) Determine the tension capacity of 150×90× 8 mm angles in Fe 410 steel assuming
 - i) Connection through the longer leg by two rows of M20 bolts
 - ii) Connection through shorter leg by a single row of M24 bolts.
 - b) What are the types of transmission line towers? What is meant by tower configuration and explain in detail the various loads to be considered in design of transmission tower?
8. Design a purlin for a roof truss having the following data:
Span of the truss = 6m
Spacing of the truss = 3m c/c
Inclination of the roof = 30
Spacing of purlin = 2m c/c
Wind pressure = 1.5kN/m²
Roof coverage = A.C sheeting weighing 200N/m²
Provide a channel section purlin.
